Point-Based Estimator for Arbitrary-Oriented Object Detection in Aerial Images

Kun Fu, Zhonghan Chang, Yue Zhang, *Member, IEEE*, and Xian Sun, *Senior Member, IEEE*

***Abstract*— Object detection in aerial images is important for a wide range of applications. The most challenging dilemma in this task is the arbitrary orientation of objects, and many deep-learning-based methods are proposed to address this issue. In previous works on oriented object detection, the regression-based method for object localization has limited performance due to the shortage of spatial information. And the models suffer from the divergence of feature construction for object recognition and localization. In this article, we propose a novel architecture, i.e., point-based estimator to remedy these problems. To utilize the spatial information explicitly, the detector encodes an oriented object with a point-based representation and operates a fully convolutional network for point localization. To improve localization accuracy, the detector takes the manner of coarse-to-fine to lessen the quantization error in point localization. To avoid the discrepancy of feature construction, the detector decouples localization and recognition with individual pathways. In the pathway of object recognition, the instance-alignment block is involved to ensure the alignment between the feature map and oriented region. Overall, the point-based estimator can be easily embedded into the region-based detector and leads to significant improvement on oriented object detection. Extensive experiments have demonstrated the effectiveness of our point-based estimator. Compared with existing works, our method shows state-of-the-art performance on oriented object detection in aerial images.**

***Index Terms*— Aerial images, convolutional network, oriented object, point-based estimator.**

Manuscript received May 5, 2020; revised July 16, 2020; accepted August 25, 2020. Date of publication September 10, 2020; date of current version April 22, 2021. This work was supported in part by the National Natural Science Foundation of China under Grant 61725105 and Grant 41801349, and in part by the Gusu Innovation Talent Foundation of Suzhou under Grant ZXT2017002. *(Zhonghan Chang and Kun Fu contributed equally to this work.) (Corresponding author: Zhonghan Chang.)*

Kun Fu is with Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, also with Aerospace Information Research Institute, Chinese Academy of Sciences, Suzhou 215000, China, and also with the Key Laboratory of Network Information System Technology (NIST), Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China.

Zhonghan Chang is with Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, also with the Key Laboratory of Network Information System Technology (NIST), Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100049, China, and also with the University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: changzhonghan16@mails.ucas.ac.cn).

Yue Zhang and Xian Sun are with Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, and also with the Key Laboratory of Network Information System Technology (NIST), Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China.

Color versions of one or more of the figures in this article are available online at https://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2020.3020165

I. Introduction

OBJECT detection in aerial images has been one of the hottest topics in the field of remote sensing and computer vision [1]–[7]. Developed from the deep convolutional networks (ConvNets) [8]–[10], many works have been proposed to advance object detection in aerial images [5], [7], [11]–[13]. In natural-scene images with horizontal perspectives, the objects have vertical dominant directions. Aerial images are typically collected with the top views, and the objects usually have arbitrary orientations. When the objects are densely packed in aerial images, the axis-aligned box will cover irrelevant background (BG) or neighbor instances [2]. This issue is a major distinction between object detection in aerial images and the task in natural-scene. To solve this issue, the works in [2], [14], [15], and [6] introduce the novel task of oriented object detection, in which the object is localized with an oriented box. This task requires the detector to estimate the exact pose of an oriented object, rather than providing the coarse position of the object merely. The challenging task motivates us to rethink the localization method and detection pipeline for the arbitrary-oriented object in aerial images

Most works [6], [14], [16]–[19] build detectors for the oriented object in aerial images upon existing methods [8], [20] which show excellent performance on the natural-scene task. For example, the FR-O [2] extends Faster R-CNN [8] to localize the four-corners of an oriented object. However, existing methods fail to localize arbitrary-oriented objects accurately due to several critical drawbacks. First, previous methods fail to fully utilize spatial information. As illustrated in Fig. 5(c), previous works utilize the regression-based method, which encodes the oriented object box as an offset vector [14], [17], [18]. The model operates fully connected layers (fc-layers) to predict the vector. On the convolutional feature, each cell responses to the neighbor region on the image [9], [21]. When the feature is collapsed into a vector, the pixel-to-pixel mapping between the feature map and image is lost. The missing of spatial information hinders the capability of the model on localizing objects [22]–[25]. Second, previous methods suffer from the discrepancy of feature construction. Toward accurate oriented box localization, the network should respond to the small translation of an object, constructing translation-covariant features. On object recognition, the feature map should be invariant to the spatial translation of the object. As illustrated in Fig. 2(a), previous methods allocate two tasks of different intentions to a union pathway, which provides suboptimal representation merely.

0196-2892 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:47:08 UTC from IEEE Xplore. Restrictions apply.

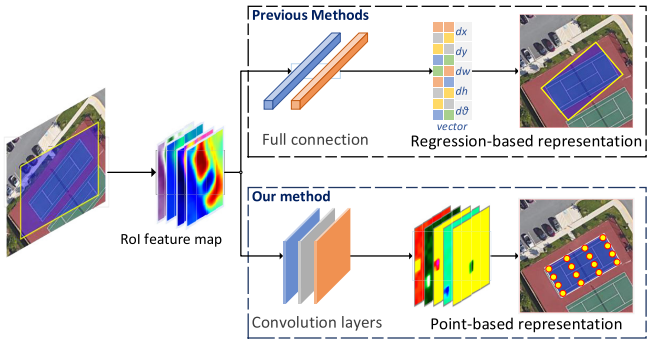


Fig. 1. Different methods for oriented object detection. 1) The previous method predicts the *regression offset (a vector)* between the proposal box and OBB. The feature loses spatial information after the fc-layers. 2) In our method, the model exploits the *point-based representation* for oriented objects. It adopts FCN to generate dense predictions for object localization. This design can preserve the spatial information of convolutional features instead of mixing the feature into a flatten vector. The changes (e.g., object rotation and scaling) in image lead to the corresponding variation on the output (e.g., rotated and scaled mesh).

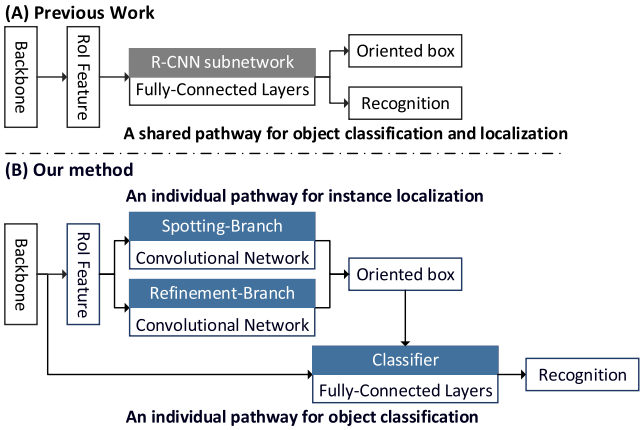


Fig. 2. Illustration of the detection pipeline. (a) In the previous works, the detector adopts a shared pathway for oriented box prediction and object classification. It has the discrepancy of learning translation-covariance for object localization and learning translation-invariance for recognition. (b) In our method, each subtask is allocated with an individual pathway. The pathway for object localization constructs translation-covariant feature maps to capture the small translations of objects. The classifier catches information from the other pathway for recognition. This design avoids the discrepancy of feature construction.

In this article, the point-based estimator is proposed to remedy these issues. 1) The model can utilize spatial information explicitly. We design a new point-based representation that localizes the layout of an object with a set of points. Based on this conception, the point-based estimator operates a fully convolutional network (FCN) to localize the points, as illustrated in Fig. 1. The model encodes the layout of an oriented object on the feature map explicitly, and the output of the network can generate covariant response to the translations of objects. Compared with the regression-based method lacking spatial information, our method has stronger capability on modeling the spatial translation of objects and improves localization accuracy. 2) To avoid the divergence of feature construction, oriented object detection is decoupled into subtasks, i.e., instance localization and object recognition. Each subtask is allocated with a separate pathway, as illustrated in Fig. 2(b). The network does not have to catch information of discrepant properties in a single pathway.

Toward accurate oriented object localization, two FCN modules (i.e., spotting-branch and refinement-branch) are exploited to localize points in the manner of coarse-to-fine. This design can avoid the quantization error in point localization compared with previous works [20], [26]. It provides the exact position of the points and promotes the localization accuracy of the oriented object consequently. Moreover, we develop a point-based estimation method to restore the bounding box of the oriented object. This design provides a mechanism to reduce the influence of inaccurate predicted points via the constraint of mesh layout. In the pathway of object recognition, the instance-alignment block is introduced to extract the features of oriented regions for the classifier. Rather than RoI-pooling or RoI-Align handling axis-align region [8], [20] merely, it provides a more general operation for extracting features from regions.

In this article, we make the following contributions:

1) We propose innovative point-based representation and detection network to advance oriented object detection in aerial images.
2) An insightful method is proposed to eliminate the divergence of feature construction. It verifies the usefulness of decoupled feature pathways to the object detector.
3) The point-based estimator exploits the coarse-to-fine fashion and the spatial information of convolutional network to improve object localization.
4) Toward accurate object recognition, the model involves the instance-alignment block to ensure the alignment property of feature extraction.

Extensive ablation studies are conducted to verify the effectiveness of our developments. Our method is compared with state-of-the-art methods and achieves competitive performance on arbitrary-oriented object detection in aerial images. In the rest of the article, we first review some related works in Section II. Then our method is described in detail in Section III. Finally, Section IV presents the ablation study, analysis and comparison with state-of-the-art methods.

II. Related Works

A. ConvNets-Based Object Detectors

1) Region-Based Detector: With the success of ConvNets, much progress has been made on object detection [8], [20], [27]–[29]. Region-based detectors have achieved excellent detection accuracy [8], [20], [29]–[31]. The Regions with convolutional neural network features (R-CNN) [31] is the first to combine the region proposal method with ConvNets. Developed from R-CNN, FR [8] first uses region proposal network (RPN) to generate region-of-interest (RoIs). Then, the model extracts feature from each RoI box via RoI-pooling (or RoI-Align) operation and operates R-CNN subnetwork for region-wise classification and bounding-box regression. The previous works [24], [32], [33] show that adding convolutional layers to the R-CNN subnetwork improves the localization accuracy. However,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:47:08 UTC from IEEE Xplore. Restrictions apply.

these detectors still use fc-layers and regression method to localize objects. The extra conv-layers are used to enhance the regional features merely. The spatial information has an implicit role in box prediction. Divergence of feature construction exists in these methods since the model utilizes a shared sequence of fc-layers for prediction.

2) Fully Convolutional Network: Different from recognition models [9], [10] dumping the spatial information via fc-layers, the FCN scans the input image and outputs dense classification maps [34]. On the output map, each cell acts as an individual classifier for the corresponding image region. On semantic segmentation, the models [22], [23], [34], [35] take advantage of the pixel-to-pixel property of FCN and localize boundaries of objects accurately. The insightful work [25] also verifies that the conv-layer has stronger ability on object localization than fc-layers. Driven by these novel points, we build object localization module with fully convolution architecture to utilize the spatial information explicitly.

3) Object Detection With Key-Points: Recently, some works [26], [36], [37] adopt key-points detection methods on the natural-scene object detection. The CornerNet [36] and ExtremeNet [37] extend single-stage detector with hourglass network [38]. The model localizes the corners or center point of objects, and groups the relative points of the same object to a bounding box. Inspired by key-points detection method[1], our model treats the oriented object as points in a mesh and develops a mechanism to calibrated localization error. Many works [20], [38], [41]–[43] were proposed to advance key-point detection. Mask R-CNN [20] is a popular key-point detector, and it appends a FCN branch for key-points to the R-CNN subnetwork. Based on Mask R-CNN, the Grid R-CNN [26] proposes to detect the grid of the object and shows improvement over FR. However, Mask R-CNN maps the key-point to the nearest cell on the mask, which involves quantization error in key-point localization[2]. To reduce quantization error, we propose a fashion of coarse-to-fine for point localization.

B. Oriented Object Detection in Aerial Images

In the field of aerial image analysis, object detection is an important topic. Most of the early methods [5], [11], [13], [44], [45] locate objects with axis-align bounding boxes, which fails to handle objects with arbitrary rotation and densely distribution. DOTA data set [2] was proposed to remedy this issue by localizing objects with oriented bounding boxes (OBB) and provides a benchmark to advance research and application on this task. On oriented object detection in aerial images, some works [14], [16], [17], [19] proposed to build models upon region-based and natural-scene detectors [8], [46]. The methods in [14], [47], [48] depend on rotated anchors to locate the oriented ships, which is more accurate than using the axis-align anchors. These methods have a large number of anchors and suffer from heavy computation cost and the imbalance issue in anchor matching [17], [49]. The RoI-transformer [17] developed FR by adding the RRoI learner to output rotation RoIs and rotated position sensitive (RPS)-RoI-Align to extract position-sensitive feature from RoIs. The CAD-Net [28] was proposed to incorporate global and local contextual from ResNet and feature pyramid network (FPN) into FR. Specifically, It exploits pyramid local context network, global context network, and spatial-and-scale-aware attention module to construct and enhance feature maps.

The mentioned works make further developments on FR by combining the characteristics of aerial image. However, these methods still use the regression method with fc-layers for bounding boxes and suffer from the shortage of spatial information. The divergence of feature construction also exists in existing methods since the methods mentioned above utilize shared pathway for object localization and classification. The existing methods have slight differences in the basic representation of oriented object. The FR-O [2] localizes an object with its four-corners. Some works [6], [16], [17], [28], [50], [51] defines oriented object with its center point, size, and angles. The R2CNN [52] localizes an object with its edge-centers. However, these methods predict each attribution of the oriented box independently and lack a strategy to restore inaccurate predictions. A method to calibrate the inaccurate points is involved in our works.

III. Methods

In this section, we first illustrate the point-based representation for an oriented object. Then, we build the point-based estimator upon the region-based detection pipeline, as illustrated in Fig. 3. The point-based estimator takes the RoIs from RPN as input and outputs category label and oriented box for each RoI. It decouples oriented object detection by operating instance localization and object recognition in sequence. Specifically, in the pathway of instance localization, the spotting-branch and refinement-branch localize the points of an oriented object in the manner of coarse-to-fine. Then, the OBB of an object is estimated according to the predicted points. In the pathway of object recognition, the instance-alignment block extracts feature from oriented boxes for the fully connected classifier.

A. Point-Based Representation for an Oriented Object

The position of an oriented object is denoted as a four-tuple $(x_i,\ y_i)_{i\in[1,\ 4]}$, as illustrated in Fig. 4(a). The tuple $(x_i,\ y_i)$ denotes the coordinate of the ith corner of the box. In the task of arbitrary-oriented object detection [2], the model should find the corners of each object and minimizes localization error.

In our method, as illustrated in Fig. 4(b), the oriented object is defined as $N \times N$ equally spaced points within the bounding box. The set of all points is denoted as $U = \{u_{ij}\}_{i,\ j\in[1,N]}$. The point $u_{ij} = (x^u_{ij},\ y^u_{ij})$ locates at the ith row of jth column in the mesh. For clarity, the set of points can also be denoted as $U = \{u_k\}_{k\in[1,K]}$ and $K = N \times N$.

B. Points Localization

The pathway of localization is exploited to predict the point-based representation of an oriented object. The modules

[1]In our method, the object is divided as sampling points with equal intervals in the bounding box. It does not need additional manual annotations. This strategy is different from the key-point detection tasks [39], [40] which labels the human body based on prior knowledge and additional manual annotations.

[2]This issue is described in detail in Section III-B.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:47:08 UTC from IEEE Xplore. Restrictions apply.

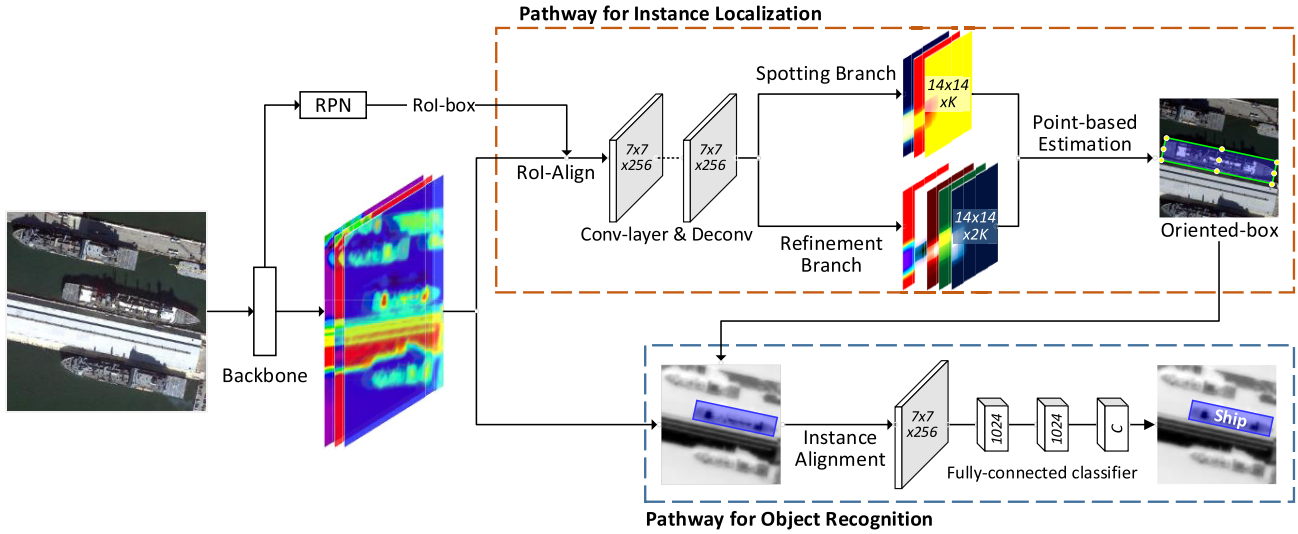


Fig. 3. Detection pipeline of the model with point-based estimator for arbitrary-oriented object. The model cascades two separate pathways for instance localization and object recognition. In the pathway of instance localization, the spotting-branch and refinement-branch are implemented with fully convolutional architecture to preserve spatial information. The pathway of object recognition is used to classify the oriented box from the localization pathway. The instance-alignment block extracts feature from the arbitrary-oriented boxes. The classifier uses fc-layers to capture the information of the entire object.

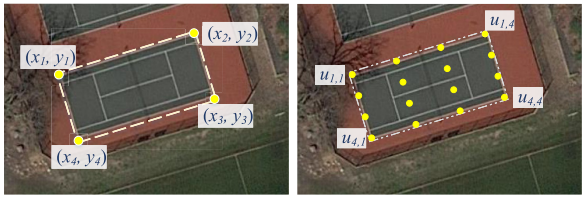


Fig. 4. Oriented box representation. (a) Given a rotated object, the coordinates of four corners are used to denote the position of an object. (b) In our method, the point-based representation of an object contains a set of points (4×4 in this figure) with equal intervals. This figure is an example of indexing the points in practice. The left corner of the upper longest-edge is denoted as the first point $u_{1,1}$. The point $u_{4,4}$ is the right corner of lower longest-edge.

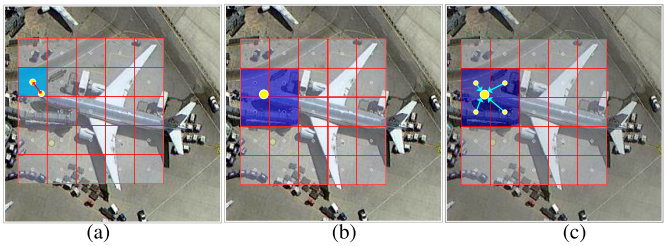


Fig. 5. Coarse-to-fine manner of point localization. (a) When the feature of RoI (*red grid*) is extracted and scaled by RoI-Align (or RoI-pooling), the point (yellow) may have fractional coordinates on the feature map of RoI. There exists localization error (*red arrow*) between the (*blue*) cell and the point. (b) In our method, the spotting-branch outputs a confidence score of a point at each position. It spots the positive (*blue*) cells surrounding the points. (c) Refinement-branch predicts the transformation (*cyan arrows*) between the point and positive cells (a) Quantization errors. (b) Spotting. (c) Refinement.

have the design of FCN to preserve spatial information and maintain translation-covariance to object. Given a RoI box covering an object, the detector operates RoI-Align to extract the feature of each RoI with a fixed size of $M_p \times M_p$, as illustrated in Fig. 3. The FCN processes the RoI feature and outputs dense prediction with the size of $M \times M$, here $M = 14$ and $M_p = 7$ by default.

The RoI box is denoted as $b = \{x, y, w, h\}$, and the parameters (x, y) and (w, h) denote the top-left corner and sizes of the box, respectively. The coordinates of points projected on the RoI feature are denoted as $P = \{p_k\}_{k\in[1,K]}$. Here, the transformation from U to P includes scaling and shift, and it is defined as

$$p_k = (x_k^p,\ y_k^p), \quad x_k^p = (x_k^u - x)\frac{M}{w},\ \ y_k^p = (y_k^u - y)\frac{M}{h}. \quad (1)$$

To localize points P, the works [20], [26] present an available but not a promising method. As illustrated in Fig. 5(a), the network is trained to predict one-hot confidence map in where only the nearest cell to the point is labeled as foreground. Specifically, the point $p_k = (x_k^p,\ y_k^p)$ is mapped to the nearest cell $(\lfloor x_k^p \rfloor, \lfloor y_k^p \rfloor)$ on the feature map. There exists localization error between the predicted cell and the point, i.e., $(x_k^p - \lfloor x_k^p \rfloor, y_k^p - \lfloor y_k^p \rfloor)$. The error limits the precision of point localization. Therefore, our model exploits a new method for point localization rather than inheriting the previous works [20], [26].

To lessen the quantization error, we propose to operate point localization in the manner of coarse-to-fine. As shown in Fig. 5(b) and (c), spotting-branch provides candidate positions by spotting the neighbor cells of the target point. The refinement-branch generates compensation between the neighbors and the exact position of the point.

1) Spotting-Branch: The spotting-branch generates dense confidence maps on each RoI, as shown in Fig. 5(b). The spotting-branch has the design of FCN, and the output layer has channels of K, as shown in Fig. 6. Here, K is the total number of point types. At the end of the network, the predicted score is normed by the sigmoid activation. The output is denoted as m, and it has a size of $M \times M \times K$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:47:08 UTC from IEEE Xplore. Restrictions apply.

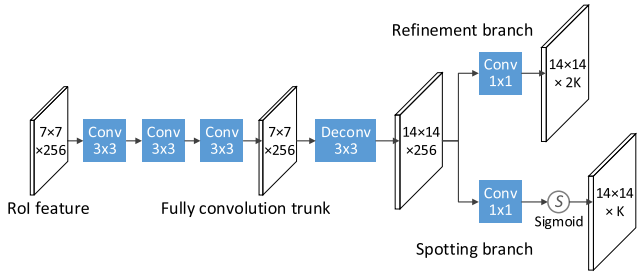


Fig. 6. Architecture of the spotting branch and refinement branch. The spotting branch is equipped with FCN and generates a $M \times M \times K$ confidence map ($14 \times 14 \times K$ by default). It has a sigmoid function. The refinement branch outputs a $M \times M \times 2K$ dense prediction. The refinement branch shares the FCN trunk with the spotting branch, which avoids adding too many parameters. The 3×3 convolutional layers have the stride of 1 and channels of 256. The 3×3 deconvolutional layer has the stride of 2 and channels of 256.

The kth channel m_k [e.g., Fig. 5(b)] denotes the confidence map for the kth type of point. The cell $m_k(v_i) \in (0, 1)$ denotes the probability of tiling the kth point at the position $v_i = (x_i^v, y_i^v)$. Here, the variable $k \in [1, K]$ is indexing the points of the object, and the variable $i \in [1, M \times M]$ is indexing the cells on the confidence map.

For training the spotting-branch, the ground-truth $\bar{m}$ is defined as a binary mask and has the size of $M \times M \times K$. The kth channel $\bar{m}_k$ denotes the ground-truth of the kth point. The cell $\bar{m}_k(v_i)$ denotes the ground-truth value at the position of v_i. If the cell v_i and the point p_k have $|x_k^p - x_i^v| \leq 1$ and $|y_k^p - y_i^v| \leq 1$, it will be assigned positive label $\bar{m}_k(v_i) = 1$, as shown in Fig. 5(b). The rest cells have negative labels $\bar{m}_k(\cdot) = 0$. The loss function of kth point is defined as the logistic loss over the output map

$$L_s(k) = \frac{1}{M \times M} \sum_{i=1}^{M \times M} (-\bar{m}_k(v_i) \log(m_k(v_i)) - (1 - \bar{m}_k(v_i)) \log(1 - m_k(v_i))). \quad (2)$$

And the loss function of spotting-branch is the average over all the point types

$$L_{\text{spot}} = \frac{1}{K} \sum_{k=1}^{K} L_s(k). \quad (3)$$

2) Refinement-Branch: The refinement-branch has the design of FCN, and the output layer has channels of $2K$, as shown in Fig. 6. The output is denoted as h, and it has the size of $M \times M$. As illustrated in Fig. 5(c), the $h_k(v_i)$ denotes the transformation between the cell v_i and the point p_k, and it is defined as a 2-D vector. At the position of v_i, the ground-truth $\bar{h}_k$ of the point p_k is defined as

$$\bar{h}_k(v_i) = p_k - v_i = \left(x_k^u - x_i^v,\ y_k^u - y_i^v\right). \quad (4)$$

Here, the refinement-branch is trained to fit the transformation $\bar{h}_k(v_i)$ between each position $\{v_i\}_{i \in [1,\ M \times M]}$ and each point $\{u_j\}_{j \in [1,\ K]}$, respectively.

For training the refinement-branch, the loss function of kth point is defined as the sum of smooth-L_1 losses [8] over the output map

$$L_r(k) = \sum_{i=1}^{M \times M} \bar{m}_k(v_i) \cdot \text{smooth}_{L1}(\bar{h}_k(v_i) - h_k(v_i)). \quad (5)$$

Here, the factor $\bar{m}_k(v_i)$ evaluates to 1 with positive cell and 0 otherwise. Only positive positions surrounding the point participate in the optimization. The loss function of the refinement-branch is defined as average over all the points

$$L_{\text{ref}} = \frac{1}{K} \sum_{k=1}^{K} L_r(k). \quad (6)$$

For each RoI box, the loss function of point localization is defined as

$$L_{\text{pt}} = u \cdot L_{\text{ref}} + u \cdot L_{\text{spot}}. \quad (7)$$

Here, the variable u denotes the label of input RoI. If a RoI has sufficient intersection over union (IoU) overlap with an object as $IoU > 0.4$, and it has positive $u = 1$. The rest RoIs covering BG or small pieces of foreground have $u = 0$. The negative RoIs are filtered in the loss function, and the model only concentrates on the positive RoIs [53], [54]. If the RoI has positive relationships with multiple objects, it is matched to the object with max overlap.

3) Inference: The point-based estimator utilizes confidence map m and refinement map h to localize the points on the RoI. At first, at each position v_i, we use the transformation $h_k(v_i)$ to localize the candidate position of point

$$\tilde{v}_{i,k} = (h_k(v_i) + v_i). \quad (8)$$

Then, we utilize the weighted-average of candidate positions $\{\tilde{v}_{i,k}\}_{i \in [1,\ M \times M]}$ to compute the coordinates of point

$$\lambda_k = \sum_{i=1}^{M \times M} m_k(v_i) \quad (9)$$

$$\tilde{p}_k = \frac{1}{\lambda_k} \sum_{i=1}^{M \times M} m_k(v_i) \cdot \tilde{v}_{i,k} \quad (10)$$

here, the predicted score $m_k(v_i)$ acts as the weight in the summation. The result $\tilde{p}_k$ denotes the predicted position of kth point, and $\tilde{p}_k = (x_k^p, y_k^p)$. The variable λ_k denotes the sum of the confidence scores and it acts as normalization factor in the summation. If there are no positive positions for the kth point, the invalid point $\tilde{p}_k$ will have inferior confidence $\lambda_k \to 0$.

The coordinates of the point $\tilde{p}_k$ are remapped from the feature map of RoI to the input image via 1. The transformed point is denoted as $\tilde{u}_k$. After the procedure of spotting and refinement, the set of predicted points is denoted as $\tilde{U} = \{\tilde{u}_k\}_{k \in [1,K]}$, and the set of confidence scores is denoted as $\Lambda = \{\lambda_k\}_{k \in [1,K]}$. The results can also be denoted as $\tilde{U} = \{\tilde{u}_{ij}\}_{i,j \in [1,N]}$ and $\Lambda = \{\lambda_{ij}\}_{i,j \in [1,N]}$ for clarity.

C. Point-Based Estimation for Oriented Bounding Box

In our method, the point-based representation of an object is predefined as equally spaced points in $N \times N$ mesh. During inference, the layout of points may be distorted by the prediction error. We design a mechanism to calibrate the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:47:08 UTC from IEEE Xplore. Restrictions apply.

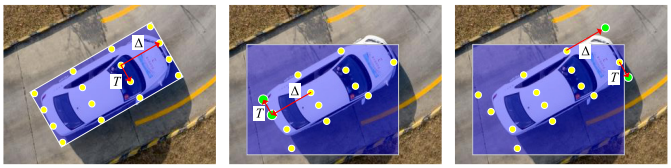


Fig. 7. Point-based estimation for the orientation box. (a) Vector Δ denotes the transformation between two neighbor points along the dominant direction of the object. The vector T has a perpendicular direction to Δ. (b) Inaccurate point (green point) is calibrated by the constraint of other points in the mesh. (c) Green point goes beyond the RoI (blue box), it is restored by the transformations (red arrows) and valid points (yellow points).

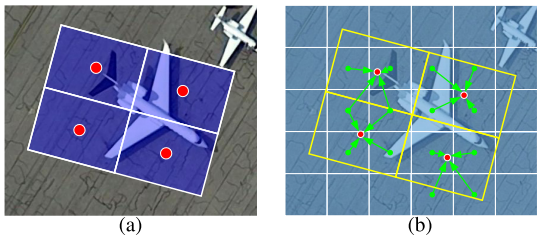


Fig. 8. Feature extraction on an oriented region. (a) Oriented box is allocated with a 2×2 sampling points. In practice, we use 7×7 grids to cover the oriented region. (b) Feature value of the sampling point (red) is computed via bilinear interpolation (green arrows) over neighbor cells.

inaccurate points in a mesh. As shown in Fig. 7, we first estimate the basic transformations between the neighbors. Then, the transformations are used to calibrate the inaccurate points in the mesh. In the estimation, the points with high scores make major contributions to the result. The points falling out of RoI scope have low scores and make inferior contributions to the result. By the ensemble of all predicted points, the overall deviation of the oriented box can be reduced.

The predicted points are denoted as $\tilde{U} = \{\tilde{u}_{ij}\}_{i,j\in[1,N]}$ and confidence scores are denoted as $\Lambda = \{\lambda_{ij}\}_{i,j\in[1,N]}$. The transformation between two neighbor points along the longest-edge of the oriented box is defined as

$$z = \sum_{i=1}^{N}\sum_{j=2}^{N} \lambda_{ij} \cdot \lambda_{i,j-1} \tag{11}$$

$$\Delta = \frac{1}{z}\sum_{i=1}^{N}\sum_{j=2}^{N} \lambda_{ij} \cdot \lambda_{i,j-1} \cdot (\tilde{u}_{ij} - \tilde{u}_{i,j-1}). \tag{12}$$

Here, z acts as the normalization factor. The factor $(\lambda_{ij} \cdot \lambda_{i,j-1})$ suppresses the invalid point with inferior score. Then, the transformation between two neighbor points along the short-edge is defined as

$$z = \sum_{i=2}^{N}\sum_{j=1}^{N} \lambda_{ij} \cdot \lambda_{i-1,j} \tag{13}$$

$$T = \frac{1}{z}\sum_{i=2}^{N}\sum_{j=1}^{N} \lambda_{ij} \cdot \lambda_{i-1,j} \cdot (\tilde{u}_{ij} - \tilde{u}_{i-1,j}). \tag{14}$$

Here, Δ and T are 2-D vectors, having $\Delta = (\delta_x, \delta_y)$ and $T = (\tau_x, \tau_y)$. The shift τ_y can also be defined as $\tau_y = -\delta_x/\delta_y \cdot \tau_z$, since Δ and T have orthogonal directions.

The top-left corner is estimated via all points and transformations

$$z = \sum_{i,j} \lambda_{ij} \tag{15}$$

$$u_{1,1} = \frac{1}{z}\sum_{i,j} \lambda_{ij}(\tilde{u}_{ij} - (j-1)\cdot\Lambda - (i-1)\cdot T). \tag{16}$$

Finally, all corners are restored via the top-left corner and transformations, and it is defined as

$$u_{ij} = u_{1,1} + (j-1)\cdot\Lambda + (i-1)\cdot T. \tag{17}$$

The results $\{u_{1,1}, u_{1,N}, u_{N,N}, u_{N,1}\}$ can denote the four corners of an OBB. As mentioned in Section III-A, only the corners of the object are required in the task of oriented object detection [2]. The rest points only participate in the preceding estimation, and they are not used to form final detection boxes. The operation in the estimation of the oriented box is simple (summation, adding and multiplying over low-dimensional tensors), and the computation cost is negligible compared with the preceding neural network.

D. Object Recognition

The pathway of object recognition takes input as the oriented detection boxes from the preceding module. For each box, it extracts the feature of the oriented region and operates recognition on the extracted feature, as illustrated in Fig. 3.

1) Instance-Alignment Block: The RoI-pooling and RoI-Align [8], [20], [30] are designed for axis-aligned RoI fails to handle the oriented region. The instance-alignment block is proposed for feature extraction on arbitrary-oriented regions. Specifically, the oriented box is treated as a set of sampling points, and the feature vector of each point is computed via interpolation operation, as shown in Fig. 8.

The backbone scans the input image and outputs feature map $F \in \mathbb{R}^{H\times W\times D}$. Here, the parameters H, W, and D denote the height, width, and channels of feature F. Given an oriented box, the output of feature extraction is denoted as $H \in \mathbb{R}^{M_a\times M_a\times D}$. The parameter M_a also controls the number of sampling points. Given an oriented box, the set of sampling points with equal intervals is denoted as $E = \{e_{ij}\}_{i,j\in[1,\,M_a]}$. Here, e_{ij} locates at the ith row and jth column of the sampling grid and $e_{ij} = (x^e_{ij},\ y^e_{ij})$. The feature vector $H(i,j)$ of the sampling point e_{ij} is defined as

$$H(i,j) = \frac{1}{z}\sum_{n=1}^{W\times H} \omega_{ij}(n)\cdot F_n,\ z = \sum_{n=1}^{W\times H}\omega_{ij}(n). \tag{18}$$

Here, the feature vector F_n denotes the value of the nth cell on the feature F, and it has coordinates of $(x^f_n,\ y^f_n)$. The parameter $\omega_{ij}(n)$ denotes the weight of bilinear interpolation between the sampling point e_{ij} and the feature cell F_n, and it is defined as

$$\omega_{ij}(n) = \max(0,\ 1 - |x^f_n - x^e_{ij}/s|) \cdot \max(0,\ 1 - |y^f_n - y^e_{ij}/s|) \tag{19}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:47:08 UTC from IEEE Xplore. Restrictions apply.

here, the parameter s denotes the scaling factor between the input image and down-sampled feature F, and it is used to scale the coordinates of points from image to the feature map.

2) Classifier and Training: The classifier is instantiated with fc-layers to capture the scope of the entire object following image classification models [9], [10]. For each input box, the classifier processes extracted feature H and generates the confidence over all categories, denoted as q. For training the classifier, if the input box has an IoU higher than 0.5 with any ground-truth box, it is allocated with a foreground class label. The rest boxes are allocated with BG labels. If the input box overlaps with several ground-truth objects, the object that has the highest IoU with the input box is treated as the target. The classification loss is defined as L_{cls}, and it adopts the cross-entropy loss [8], [9] over all categories. For each RoI box, the total loss function of point-based estimator follows the way of multitask learning [3], and it is defined as a summation operation

$$L_{\text{kp}} = L_{\text{cls}} + L_{\text{pt}}. \tag{20}$$

Here, the loss L_{pt} for point localization is illustrated in 7.

E. Network Architecture

The hyperparameter in Sections III-B and III-D are assigned $M_p = 7$, $M_a = 7$. It is analogous to FR extracting 7×7 features from RoIs. By default, the number of points in the point-based estimator is set as $N = 4$. In the pathway of instance localization, it has three 3×3 convolutional layers (conv-layers) and one deconvolution layer (deconv-layer) [20], [34] followed by two 1×1 conv-layers, as shown in Fig. 6. The deconv-layer has the stride of 2 and up-samples the RoI feature size from 7×7 to 14×14. In the ending of pathway, the first 1×1 conv-layer has output channels of $2K$ and serves as the refinement-branch. The other 1×1 conv-layer serves as the spotting-branch. It has output channels of K and ends with a sigmoid function. In the pathway of recognition, the classifier stacks two fc-layers with channels of 1024 and one output layer.

The options of the backbone include ResNet [46] and FPN [55]. The detection pipelines are implemented as follows:

1) *Vanilla ResNet*: the RPN connects to the conv-4x stage of ResNet. An extra conv-layer is appended to the last layer of ResNet and reduces the channels of feature map from 2048 to 256 [8], [56]. The R-CNN subnetwork connects to the dimension-reduction layer and the RPN.
2) *ResNet+FPN*: ResNet is extended with a top-down architecture, building a feature pyramid for multiscale object. The RPN and R-CNN subnetwork operate on the feature pyramid, and each level of the feature pyramid is responsible for the objects of particular scales [20], [55].

In the ablation study, multiple object detectors are built upon vanilla ResNet/FPN. The R-CNN subnetwork is instantiated with the point-based estimator or other variants, as illustrated in Fig. 9. The RPN has three aspect ratios $(1:2, 1:1, 2:1)$ and five scales $(32, 64, 128, 256, 512)$ [17], [55]. The RoIs are filtered by nonmaximum suppression (NMS) with the overlap of 0.7. The RoIs of top-600 ranking are fed to the R-CNN subnetwork.

[3]Equation 20 can be instantiated with weighted-sum operation, see Appendix V-C

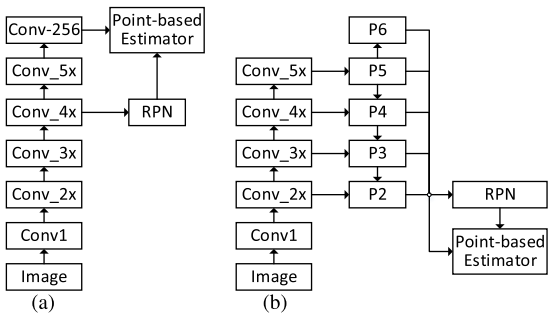


Fig. 9. Variants of object detector with different backbones. (a) Vanilla ResNet (b) ResNet + FPN.

IV. Experiments

A. Data Sets

1) DOTA: To demonstrate the effectiveness of our method, experiments are conducted on the open-public DOTA data set [2]. This article concentrates on the OBB detection task in DOTA [2]. DOTA provides a large-scale data set for oriented object detection in aerial images. It covers 188 282 instances over 15 categories. The size of images ranges from 800×800 pixels to 4000×4000 pixels. The training set has 1411 images, and the validation set has 458 images. The testing set of DOTA has 937 images. In the ablation experiments, we train the network on the training set; the evaluation is conducted on the validation set. For the comparison with state-of-the-art object detectors [2], [17], [28], the network is trained on training set and validation set together; the evaluation is conducted on the testing set.

2) HRSC2016: The high-resolution optical satellite images with complex background (HRSC) data set provides a benchmark for arbitrary-oriented ship detection in remote sensing images [48]. It collects 1061 images from Google Earth. The image size ranges from 300×300 to 1500×900. The training and validation set have 436 images and 181 images separately, and the testing set has the rest 444 images. The model is trained on the training and validation set together. We conduct evaluations on the testing set and present the comparison with state-of-the-art object detectors.

B. Implementation Details

The network is end-to-end trained using stochastic gradient descent (SGD) method [57]. The learning rate is 0.0005 for the first 200k iterations and decays to 0.00005 for the next 100k iterations. The RPN and point-based estimator participate in optimization jointly. For training the RPN, 256 anchors are selected to participate in optimization. For training the point-based estimator, we sample 512 RoIs from all proposals in one image. Aerial images have larger scales than the natural-scene images. The sliding-window method is used to solve this issue. During training, we adopt a window of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:47:08 UTC from IEEE Xplore. Restrictions apply.

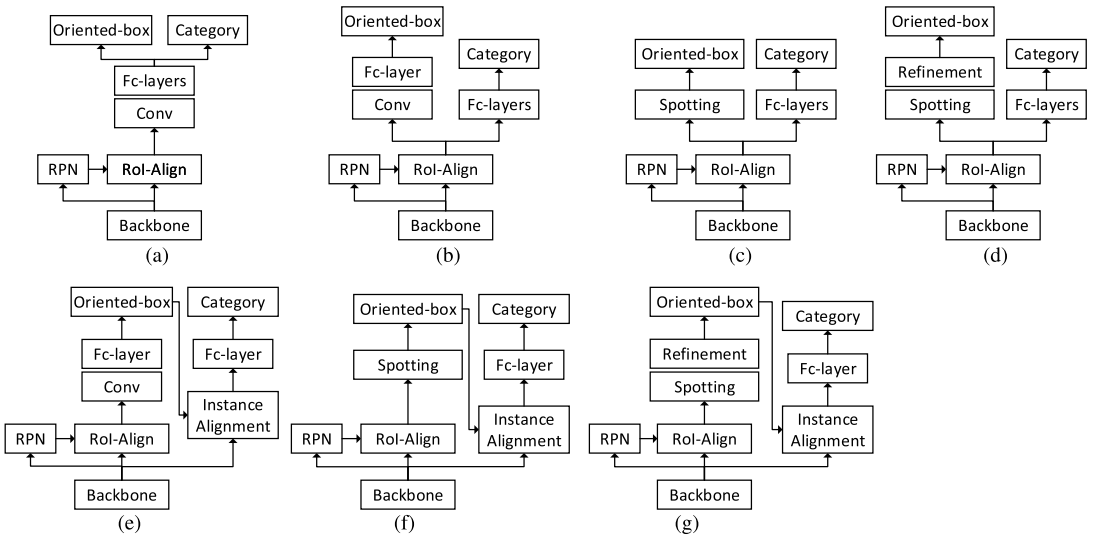


Fig. 10. Detection pipeline of variants with different settings. (a) FR-R. (b) FR-R^{dep}. (c) FR-R*. (d) FR-KP. (e) FR-Align. (f) FR-Est*. (g) FR-Est.

1000×1000 on the raw images to crop patches. The detector is trained on the patches instead of raw images.

During inference, a window of 1000×1000 slides on the input image to crop patches, and object detector processes each patch, respectively. The outputs of patches are merged to make up the detection results of whole images. NMS with IoU threshold of 0.3 is utilized to filter overlapping boxes. Here, the IoU is defined as the intersection area over union area of two oriented boxes in NMS operation [50], [52].

C. Evaluation Metrics

The overlap between oriented boxes is used to discriminate detection results [2], [40], [58]. The precision-recall curve (PRC) and average precision (AP) are used to evaluate the performance of an object detector.

1) *IoU:* The evaluation follows the protocols of oriented object detection task in DOTA [2]. The IoU is defined as the intersection area over the union area of two oriented boxes rather than axis-aligned boxes [2]. If the IoU between a detection box and a ground-truth is higher than the predefined threshold, the detection box is labeled as true-positive (TP), otherwise false-positive (FP). If a ground-truth box has no matching detections, it is labeled as false negative (FN).
2) *PRC:* After calculating the number of TP, FP and FN, the precision and recall are computed as

$$\text{precision} = \frac{\text{TP}}{(\text{TP}+\text{FP})} \tag{21}$$

$$\text{recall} = \frac{\text{TP}}{(\text{TP}+\text{FN})}. \tag{22}$$

   By computing precision and recall at every step in the ranked sequence of detections, a PRC can be drawn.
3) *AP:* AP is the area under the PRC for each class. The mean average precision (mAP) is the mean of APs over all classes. In the ablation study, the $\text{mAP}_{0.5}/\text{mAP}_{0.7}$ is computed under the IoU threshold of 0.5/0.7, respectively, [59].

D. Ablation Experiments

To investigate the performance of our methods, we conduct experiments from the baseline models to different variants of the point-based estimator. Without specific notes, ResNet-50 pretrained on ImageNet data set serves as the backbone network [46], [60]. Several models are built upon the fashion of *vanilla ResNet* (illustrated in Section III-E). The variants of object detector are illustrated in Fig. 10. The details are listed as follows:

1) *FR-R :* The work in [52] proposes a variant of R-CNN subnetwork for oriented object detection, and this design is called "rotation-module" in the following pages. The rotation-module takes an RoI box as input and generates regression offset from axis-aligned RoI to oriented box. In the FR-R, the vanilla R-CNN subnetwork is replaced by the rotation-module. To exclude the influence of additional parameters, the rotation-module cascades four 3×3 conv-layers, two fc-layers, and output layer in sequence, having the same amount of parameters with the point-based estimator.
2) *FR-R_{dep}:* It decouples classification and localization with separate pathways. The rotation-module adopts three 3×3 conv-layers and one fc-layer layer to generate offset for oriented boxes, and uses two fc-layers to predict the classification scores in parallel.
3) *FR-Kp*:* It has a simplification of the point-based estimator. The spotting-branch is utilized to localize points for oriented box estimation. The refinement-branch and instance-alignment block are removed. In parallel to the spotting-branch, RoI-Align and fc-layers process the RoI box for object recognition.
4) *FR-Kp:* The FR-Kp extends FR-Kp* by appending the refinement-branch to the spotting-branch, which removes the quantization error in point-localization. The FR-Kp and FR-Kp* have the same design of object recognition pathway.
5) *FR-Align:* The FR-Align extends the FR-R_{dep} by adding the instance-alignment block for feature extraction.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:47:08 UTC from IEEE Xplore. Restrictions apply.

TABLE I
PERFORMANCE OF BASELINE MODEL AND OUR METHODS

Method	Rotation-module	Spotting-branch	Refinement-branch	Instance-alignment	$\text{mAP}_{0.5}\%$	boost	$\text{mAP}_{0.7}\%$	boost
FR-R	✓				60.61	–	32.14	–
FR-R$_{\text{dep}}$	✓				61.75	+1.14	33.86	+1.72
FR-Kp*		✓			63.78	+3.17	35.72	+3.58
FR-Kp		✓	✓		66.59	+5.98	39.53	+7.39
FR-Align	✓			✓	62.27	+1.66	33.03	+0.89
FR-Est*		✓		✓	65.43	+4.82	36.83	+4.69
FR-Est		✓	✓	✓	68.37	+7.76	41.55	+9.41

TABLE II
PERFORMANCE OF VARIANTS WITH DIFFERENT NUMBER OF POINTS

Method	Rotation-module	Spotting-branch	Refinement-branch	$N \times N$	$\text{mAP}_{0.5}\%$
FR-R	✓				60.61
FR-Est*		✓		2×2	64.09
FR-Est*		✓		4×4	65.43
FR-Est*		✓		6×6	66.09
FR-Est*		✓		8×8	66.21
FR-Est		✓	✓	4×4	68.37
FR-Est		✓	✓	6×6	69.24
FR-Est		✓	✓	8×8	69.39

Compared with the FR-R using axis-aligned RoI box, FR-Align removes the misalignment between the RoI and oriented object.

6) *FR-Est*:* The FR-Est* extends FR-Kp* by stacking the instance-alignment block with the classifier.
7) *FR-Est:* It has all developments, and it is identical to the method in Fig. 3. The spotting-branch and refinement-branch operate on the RoI for oriented box prediction. The instance-alignment block and classifier process the oriented box for object recognition.

For a fair comparison, the models share the same design of backbone and RPN and have similar amount of parameters in practical implementation. The only difference among them is the structure of the R-CNN subnetwork.

1) Influence of Proposed Components: To explore the influence of each proposed component, we conduct experiments on the FR-R and variants of our method. The result is shown in Table I. The FR-R with rotation-module FR-R shows performance of 60.61% $\text{mAP}_{0.5}$ and 32.14% $\text{mAP}_{0.7}$ on the validation set. The FR-R_{dep} shows better performance of 61.75% $\text{mAP}_{0.5}$ and 33.86% $\text{mAP}_{0.7}$ on the validation set. It justifies the preliminary conception of decoupling recognition and localization with separate pathways.

Spotting-branch: Compared with the FR-R and FR-R_{dep} using regression-based method and fc-layers for box localization, FR-Kp* uses spotting-branch to locate objects and makes use of the spatial information explicitly. The FR-Kp* achieves boost of 3.17% $\text{mAP}_{0.5}$ and 3.58% $\text{mAP}_{0.7}$ over the FR-R baseline. The FR-Kp* also shows significant improvement over the FR-R_{dep}. The result verifies the effectiveness of using the point-base method to localize oriented objects. It also demonstrates that utilizing the spatial information explicitly helps the network to generate high-quality boxes.

Refinement-branch: Compared with the FR-Kp* having spotting-branch only, the FR-Kp with extra refinement-branch shows performance boost of 2.81% $\text{mAP}_{0.5}$ and 3.81% $\text{mAP}_{0.7}$ on performance. This result shows that operating point localization in the manner of coarse-to-fine improves the accuracy of oriented detection boxes. It also verifies that the quantization error of mapping point to the nearest feature cell does hinder the performance of the object detector.

Instance-alignment branch: After inserting the instance-alignment block, the FR-Align shows improvement of 1.66% $\text{mAP}_{0.5}$ over the FR-R. It highlights the usefulness of aligning the region feature map with the oriented box. Adding the instance-alignment block also shows consistent improvement on the point-based models. Compared with the FR-Kp*/FR-Kp, the FR-Est*/FR-Est shows improvement of 1.65% $\text{mAP}_{0.5}$/1.78% $\text{mAP}_{0.5}$, respectively.

The FR-Est has all advanced developments, and it shows improvement of 7.76% $\text{mAP}_{0.5}$ and 9.41% $\text{mAP}_{0.7}$ compared with the FR-R baseline. Consequently, adding each component of the point-based estimator always improves the performance of the object detector. The experimental result verifies the effectiveness of our method on oriented object detection in aerial images.

2) Influence of Points Number: In the point-based estimator, the number of predefined points controls the precision of point localization and complexity of prediction task. By defining more points for an object, the object detector can have more opportunities to restore the inaccurate prediction or missing points. However, adding too many points to the mesh increases the complexity of the task, and the network may have insufficient representation power to handle the problem.

To investigate the number of points, we conduct experiments on the variants of FR-Est*/FR-Est with varying settings. Table II presents the experimental result. The models have the same design except for the number of points. Having the spotting-branch only, the FR-Est* has consistent improvement over the FR-R baseline with the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:47:08 UTC from IEEE Xplore. Restrictions apply.

TABLE III
MODELS WITH HEAVY HEAD ON THE VALIDATION DATA SET

Method	Rotation-module	point-based estimator	$mAP_{0.5}$%	boost	Inf time (ms/im)
FR-R	2×fc		59.73	–	84
FR-R	4×conv256 + 2×fc		60.61	0.88	93
FR-R	4×conv512 + 2×fc		60.74	1.01	104
FR-R	8×conv256 + 2×fc		60.76	1.03	106
FR-Est*		4×conv256 + 2×fc	65.43	–	101
FR-Est*		4×conv512 + 2×fc	66.81	1.38	113
FR-Est*		8×conv256 + 2×fc	66.94	1.51	117
FR-Est		4×conv256 + 2×fc	68.37	–	102
FR-Est		4×conv512 + 2×fc	69.92	1.55	115
FR-Est		8×conv256 + 2×fc	70.05	1.68	119

increasing number of points. The model with $N = 8$ achieves the best performance of 66.21% $mAP_{0.5}$ along the variants with spotting-branch. However, compared with models with $N = 6/N = 4$, the model with $N = 8$ shows limited improvement of 0.12%/0.78% $mAP_{0.5}$. The performance of FR-Est* becomes saturated with under the setting of $N > 4$. Considering that the localization ability of the network is limited, increasing more points touches the upper-boundary of performance and will bring slight improvement to the model.

3) Depth and Width of Subnetwork: As mentioned above, the rotation-module in FR-R consists of several stacked conv-layers and fc-layers. The work [32] points that adding conv-layers to the R-CNN subnetwork improves the localization accuracy of the object detector. However, the network still uses regression-based method and fc-layers for object localization, and the conv-layers only have an implicit role in the pipeline. In the point-based estimator, the spotting-branch and refinement-branch are instantiated with FCN architecture. Each cell on the feature map is responsible for the localization task on the corresponding image region. The model utilizes conv-layers and spatial information explicitly. Consequently, the conv-layers play different roles in the rotation-module and point-based estimator.

In both rotation-module and point-based estimator, the depth and width of the subnetwork control the capability of modeling object information. To present an insight into the two pipelines, experiments are conducted by varying the FCN in the network. The result is reported in Table III[4]. For the FR-R with rotation-module, the variant without extra conv-layers shows performance of 59.73% $mAP_{0.5}$.

Adding four extra conv-layers to the subnetwork brings slight improvement of 0.88% $mAP_{0.5}$ to the FR-R. Increasing the depth of subnetwork from four layers to eight layers brings slight boost of 0.13% $mAP_{0.5}$ to the object detector. Increasing the channels of FCN to 512-d also gains minor improvement of 0.15% $mAP_{0.5}$. When the depth and width of the network increase, the FR-R becomes saturated on performance quickly. It verifies that the regression-based fashion in the rotation-module fails to fully utilize the capability of ConvNets on extracting spatial information.

For the point-based estimator, the FR-Est with four conv-layers shows performance of 68.37% $mAP_{0.5}$. When the depth of subnetwork is increased from four layers to eight layers, the model gains improvement of 1.55% $mAP_{0.5}$. Increasing the depth of subnetwork from four layers to eight layers improves the object detector with 1.68% $mAP_{0.5}$. Increasing the channels of FCN to 512-d also gains improvement of 1.55% $mAP_{0.5}$. When the depth and width of the network are increased, the point-based estimator always has improvement and gains consistent boost on the capability of modeling spatial information. It verifies that the point-based estimator makes full use of the spatial information of convolutional features.

4) Auxiliary Loss of Point Localization: Table III shows that the point-based estimator learns richer spatial information than the FR-R. Considering that the point-based estimator connects to the backbone, the training loss of point localization may also improve the representation capability of the backbone. We investigate the influence of point-based estimator to other modules in the object detector. The experimental result is reported in Table IV. The FR-R* is built upon the FR-R by adding the point-based estimator in parallel to the rotation-module. In the training time, the FR-R* operates both point-based estimator and rotation-module on the RoIs and receives supervision from both modules. During inference, the FR-R* only operates rotation-module on the RoIs, sharing the same amount of computation with FR-R.

As illustrated in Table IV, adding the proposed components to the FR-R* model gradually results in consistent improvements of 0.62%/1.13%/1.36% $mAP_{0.5}$ over the vanilla FR-R. The experimental result shows that the auxiliary loss improves the representation capability of the backbone, and the enhanced backbone improves the following rotation-module on oriented object detection. The works on instance-segmentation [20], [24] have similar experimental results. The additional loss of mask prediction leads to improvement on box detection in Mask R-CNN. In the instance-segmentation task, the ground-truth masks come from additional annotations and manual works. In our works,

[4]In Table III, for the FR-R, the tag "4×conv256 + 2×fc" means that the rotation-module stacks 4 conv-layers with 256 channels and 2 fc-layers. For the FR-Est* and FR-Est, the tag "4×conv256 + 2×fc" means that the pathway for instance localization stacks 3 conv-layers with 256-d and one deconv-layer with 256-d before the final output layers; the pathway for classification has two fc-layers. The rest abbreviations in Table III follow this rule.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:47:08 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
INFLUENCE OF AUXILIARY LOSS ON OBJECT DETECTOR

Method	Rotation-module	Spotting-branch	Refinement-branch	Instance-alignment	$mAP_{0.5}\%$	boost
FR-R	2×fc				59.73	–
FR-R*	2×fc	✓			60.35	0.62
FR-R*	2×fc	✓	✓		60.86	1.13
FR-R*	2×fc	✓	✓	✓	61.09	1.36

TABLE V
MODELS AND BASELINE ON THE VALIDATION DATA SET

Method	backbone	FPN	$mAP_{0.5}\%$	Inf time(ms/im)
FR-R	ResNet-50		60.61	93
FR-Est	ResNet-50		**68.37**	102
FR-R	ResNet-101		62.41	116
FR-Est	ResNet-101		**70.75**	124
FR-R	ResNet-101	✓	64.32	132
FR-Est	ResNet-101	✓	**72.32**	141

TABLE VI
MODELS AND BASELINE ON TESTING DATA SET

Method	backbone	FPN	$mAP_{0.5}\%$
FR-R	ResNet-50		62.32
FR-Est	ResNet-50		**70.07**
FR-R	ResNet-101		63.97
FR-Est	ResNet-101		**72.71**
FR-R	ResNet-101	✓	65.25
FR-Est	ResNet-101	✓	**74.20**

the points are defined as the equal-interval mesh within the OBB. It provides a strategy to introduce more supervision to oriented object detection without extra annotations.

5) Detector With Different Backbones: We build the detectors with ResNet-101 and FPN. The result on the validation set is reported in Table V. When the detector is built upon a deeper backbone, the improvement of point-based estimator over the baseline is still significant. Using ResNet-101 as backbone, our method achieves improvement of 8.3% $mAP_{0.5}$ over the FR-R. Adopting the deeper backbone and feature pyramid method, the model with the point-based estimator achieves the best performance of 72.32% $mAP_{0.5}$ among all models. The result demonstrates that the point-based estimator is complementary to the developments on the backbone design and multiscale architecture [46], [55]. It is worth noting that FR-Est with ResNet-50 shows superior performance with the FR-R using ResNet-101. The point-based estimator behaves improvement over the regression-based method even if the latter one uses a deeper backbone.

To conduct further comparisons, the networks are trained on the training and validation sets and evaluated on the testing data set[5]. Table VI presents the experimental result. The performances of models are improved by large margins via adding the point-based estimator. The comparison result is consistent with the results on the validation data set. The result verifies the effectiveness and robustness of point-based estimator on oriented object detection in aerial images.

6) Computational Cost: This section presents the computational complexity analysis between the point-based estimator and the baseline model. The evaluation is conducted on one NVIDIA 2080TI GPU. We report the inference time for processing one 1000 × 1000 image patch. The result is the average time cost over 5000 tests. Table III presents the computational cost of models with different settings of subnetworks. The result shows that adding the depth or width of the subnetwork increases the inference time of the model. This phenomenon appears in both the FR-R baseline and point-based estimator.

Table V presents the inference time of the models under different backbones. Under the same backbone, the point-based estimator spends a little more inference time than the FR-R baseline. The additional time cost comes from the design of decoupling pathways[6]. However, the additional time cost accounts for a tiny percentage in the entire inference latency e.g., using ResNet-101 as backbone, the FR-Est only increases 8 ms over the baseline of 116 ms. It is worth noting that the FR-Est with ResNet-50 outperforms the FR-R using ResNet-101 (68.37% mAP versus 62.41% mAP) while maintaining lower inference time (102 ms vs. 116 ms). The result demonstrates that the point-based estimator can maintain efficiency in practical application.

E. Detailed Analysis of Oriented Object Detector

1) Analysis of Localization Precision: To investigate the localization capability of different methods, we conduct analysis on the baseline FR-R and FR-Est with 4 × 4 points. The localization error between the ground-truth and prediction are collected over the validation. Here, the localization error is defined as the distance between the corners of the detection box and the corners of ground-truth, and the value is normed by the object size. The distribution of localization error is presented in Fig. 11. The metrics include means of localization error along horizontal axis and vertical axis (μ_x and μ_y) and standard deviations of localization error (δ_x and δ_y) along dominant axes. As presented in Fig. 11(a) and (b), the FR-Est reduces the standard deviation of localization error compared with the FR-R. It shows that the point-based estimator has stronger capability on oriented object localization.

Section III-C provides a mechanism to restore the inaccurate points. The model estimates the corners of the object according

[5]The extra training data brings the gap between Tables VI and V.

[6]The FR-R baseline operates object localization and classification in parallel. The point-based estimator operates the pathway for object localization and the pathway for classification in sequence.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:47:08 UTC from IEEE Xplore. Restrictions apply.

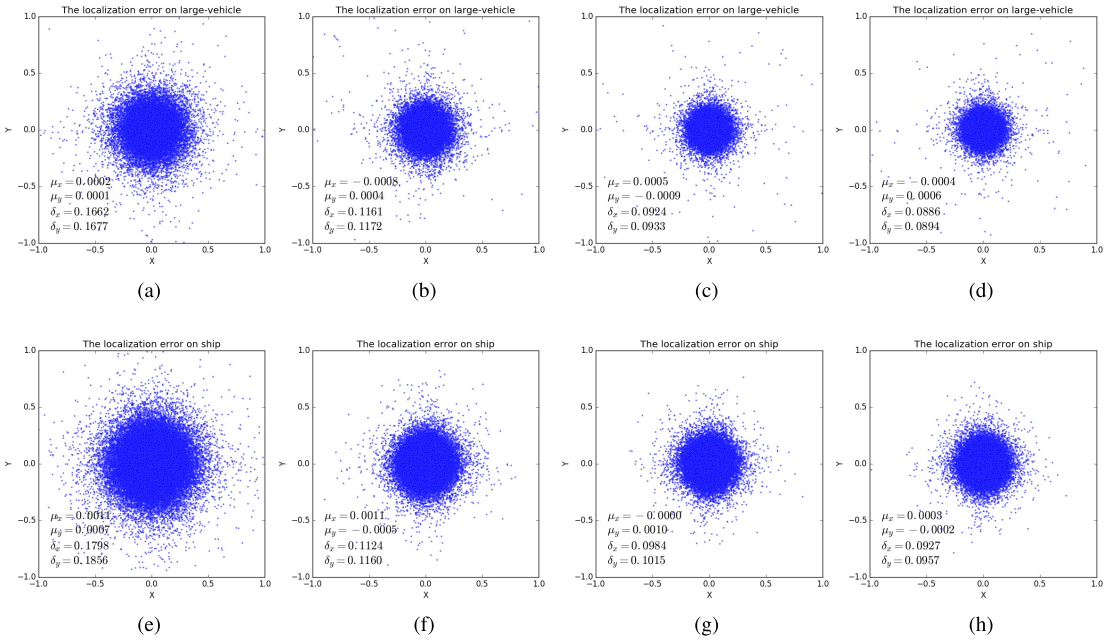


Fig. 11. Distribution of localization errors of different methods. The analysis over ship and vehicle is presented. In this figure, the variables μ_x and μ_y denote the means of localization error along the horizontal-axis and vertical-axis, respectively; the variables δ_x and δ_y denote the standard deviations of localization error along the horizontal-axis and vertical-axis, respectively. If the model has stronger capability on oriented object localization than the baseline model, it will have smaller standard deviations of a localization error than the baseline. (a) and (e) FR-R suffers from limited capability and has a large localization error than FR-Est. (b) and (f) Box estimation is not involved here. We get the predicted corners from the spotting-branch and refinement-branch directly. Without the estimation step, the FR-Est already shows a smaller localization error on corners than FR-R. (c) and (g) FR-Est also has an inferior localization error on the inner corners. (d) and (h) With the oriented box estimation, the FR-Est further reduces the localization error of predicted points. (a) FR-R on corners. (b) FR-Est on corners. (c) FR-Est on inner points. (d) Restored corners. (e) FR-R on corners. (f) FR-Est on corners. (g) FR-Est on inner points. (h) Restored corners.

to all predicted points. As shown in Fig. 11(b) and (d), the corners of the vehicle have smaller standard deviation of localization error after the estimation step. It verifies the usefulness of augmenting the definition of the oriented object from box to a set of points. The analysis result also verifies that the strategy of oriented box estimation has strong effectiveness.

2) Analysis of FP: Oriented object detection involves two child-tasks: localizing an object with a bounding box and recognizing the category of the object. To separately inspect the influence of the proposed method on the two child-tasks, we conduct analysis on the errors of object detector following previous works [32], [64]. The detection errors can be divided into two types: localization error and recognition error. Following the conception, the FPs can be divided into different types:(a)

1) *Localization Error (Loc):* If a FP sample is correctly recognized but has minor overlap ($0.1 < \text{IoU} < 0.5$) with ground truth, it is treated as localization error.
2) *Confusions With a Dissimilar Category (Oth):* If an FP sample is not Loc and it has ($\text{IoU} > 0.1$) overlap with an object of other categories, it is labeled as a confusion with a dissimilar category.
3) *Confusions With BG):* If an FP sample has insufficient overlap ($\text{IoU} < 0.1$) with any object, it is treated as a confusion with BG.

For each category of the data set, the top-ranked detections are divided into Cor[7], Loc, Oth and BG. The total number of top-ranked samples is equal to the total number of the object from the corresponding category. The proportion of true positives and FPs over top-ranked samples can be estimated. The analysis result is reported in Table VII. Experiment involves multiple models i.e., FR-R, FR-R*, FR-Est,* and FR-Est.

Compared with the FR-R baseline, the FR-Est* shows significant improvement of 3.94% Cor and reduces the localization error from 13.36% to 10.35 %. The FR-Est* shows inferior localization errors on most categories than the FR-R baseline. The experiment verifies that the point-based method can improve the localize precision of detection model on oriented objects. After adding instance-alignment block, the FR-Est shows consistent improvement of 0.82% Cor over the FR-Est*. Compared with the FR-R and FR-Est*, the FR-Est has lower recognition errors of 3.54% Oth and 11.48% BG. It verifies the importance of keeping alignment between the region-wise feature and the oriented object.

F. Comparison With State-of-the-Art Methods

To validate the effectiveness of our method, experiments are conducted on the DOTA testing set. The comparison with other state-of-the-art methods is shown in Table VIII. Considering that the detectors [6], [17], [27], [28], [61] utilize ResNet-101 and FPN [55] as backbone, we also utilize ResNet-101 and FPN to build the point-based estimator for fair comparison. Apart from our method, other methods in Table VIII still take the stereotype of regression-based method and fc-layers to localize oriented objects.

[7]Here, the TP sample is treated as Cor (correct).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:47:08 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
ANALYSIS OF OBJECT DETECTORS IN THE VALIDATION DATA SET

Metric %	Method	Ave	Plane	BD	Bridge	GTF	SV	LV	Ship	TC	BC	ST	SBF	RA	Harbor	SP	HC
Cor	FR-R	69.98	88.08	71.77	45.28	67.18	61.45	74.05	86.4	89.45	70.16	77.96	65.52	65.85	69.98	63.93	52.78
	FR-Est*	73.92	93.88	78.47	50.0	72.52	65.56	78.24	89.1	94.05	74.19	78.12	70.11	65.85	76.85	67.76	54.17
	FR-Est	**74.74**	**94.49**	**78.95**	**52.59**	**73.28**	**67.31**	**78.71**	**89.74**	**94.18**	**74.19**	**78.87**	**70.11**	**66.46**	**78.74**	**69.4**	**54.17**
Loc	FR-R	13.36	6.73	11.96	32.31	8.4	17.27	12.67	4.09	4.19	8.87	9.52	14.94	16.46	21.65	14.75	16.67
	FR-Est*	10.35	2.12	6.7	28.54	4.58	13.99	9.32	1.93	0.54	5.65	9.42	11.49	16.46	16.17	11.75	16.67
	FR-Est	**10.23**	**2.00**	**6.7**	**28.07**	**4.58**	**13.65**	**9.22**	**1.81**	**0.54**	**5.65**	**9.31**	**11.49**	**16.46**	**15.84**	**11.48**	**16.67**
Oth	FR-R	4.13	1.88	4.31	5.19	4.58	5.21	4.8	1.44	1.49	7.26	2.62	8.05	2.44	2.86	3.01	6.94
	FR-Est*	3.87	1.31	3.83	4.72	4.58	4.81	4.38	1.18	1.08	7.26	2.62	8.05	2.44	2.18	2.73	6.94
	FR-Est	**3.54**	**1.02**	**3.83**	**3.54**	**4.58**	**3.95**	**4.15**	**0.87**	**1.08**	**7.26**	**2.25**	**8.05**	**2.44**	**1.26**	**1.91**	**6.94**
BG	FR-R	12.50	3.31	11.96	17.22	19.85	16.07	8.48	8.07	4.87	13.71	9.9	11.49	15.24	5.52	18.31	23.61
	FR-Est*	11.84	2.69	11.0	16.75	18.32	15.64	8.06	7.79	4.33	12.9	9.84	10.34	15.24	4.79	17.76	22.22
	FR-Est	**11.48**	**2.49**	**10.53**	**15.8**	**17.56**	**15.09**	**7.92**	**7.58**	**4.19**	**12.9**	**9.58**	**10.34**	**14.63**	**4.16**	**17.21**	**22.22**

TABLE VIII
PERFORMANCE OF RELEASED METHODS ON DOTA TESTING DATA

Method	mAP%	Plane	BD	Bridge	GTF	SV	LV	Ship	TC	BC	ST	SBF	RA	Harbor	SP	HC
FR-Est-FPN	**74.20**	89.63	**81.17**	**50.44**	**70.19**	**73.52**	**77.98**	**86.44**	90.82	**84.13**	**83.56**	**60.64**	**66.59**	**70.59**	**66.72**	**60.55**
FR-Est	72.71	89.49	80.32	48.45	69.70	68.87	72.57	85.97	90.81	83.24	83.13	60.63	66.43	68.16	65.05	57.77
CAD-Net [28]	69.9	87.8	82.4	49.4	73.5	71.1	63.5	76.7	90.9	79.2	73.3	48.4	60.9	62.0	67.0	62.2
RCR-CNN [27]	69.47	**89.93**	77.66	44.01	62.77	67.00	69.80	79.57	90.85	80.19	78.64	58.82	56.50	68.57	63.64	53.03
FFA-3 [6]	68.9	88.8	74.4	48.8	57.9	63.6	75.9	79.6	90.8	80.3	82.9	54.3	60.0	66.9	66.8	42.5
CAD-Net [28]	67.4	88.3	71.7	51.4	66.5	72.4	64.5	76.7	90.8	77.3	74.2	45.9	60.2	65.7	56.7	48.3
LosSLPR [27]	66.95	88.92	76.88	42.13	57.18	67.21	64.98	78.36	**90.86**	79.22	78.45	51.94	56.25	57.79	61.87	52.08
FPN-RPN-O [6]	66.7	88.8	73.5	48.0	52.4	63.6	71.9	78.7	89.9	80.9	81.3	46.8	58.2	64.6	56.0	46.5
PANet [27], [24]	64.15	89.45	65.69	47.74	63.56	50.99	57.95	69.86	90.82	70.75	77.00	57.39	54.67	58.57	55.99	51.67
DFPN [61]	62.29	81.25	71.41	36.53	67.44	61.16	50.91	56.6	90.67	68.09	72.39	55.06	55.6	62.44	53.35	51.47
RRPN [17], [50]	61.01	80.94	65.75	35.34	67.44	59.92	50.91	55.81	90.67	66.92	72.39	55.06	52.23	55.14	53.35	48.22
R2CNN [17], [52]	60.67	88.52	71.2	31.66	59.3	51.85	56.19	57.25	90.81	72.84	67.38	56.69	52.84	53.08	51.94	53.58
FR-O [2]	54.13	79.42	77.13	17.7	64.05	35.3	38.02	37.16	89.41	69.64	59.28	50.3	52.91	47.89	47.4	46.3
Detectors with multi-scale training&testing																
FR-Est-FPN*	**77.62**	89.77	**85.16**	**55.32**	**76.47**	**80.08**	**83.26**	**87.53**	90.61	**87.36**	**86.89**	**65.09**	**68.48**	**71.63**	**79.6**	**66.1**
FFA-3++ [6]	75.7	**90.1**	82.7	54.2	75.2	71.0	79.9	83.5	90.7	83.9	84.6	61.2	68.0	70.7	76.0	63.7
FFA-3+ [6]	71.8	89.6	76.4	48.2	58.9	67.2	76.5	81.4	90.1	83.3	83.4	55.7	60.2	73.2	66.7	64.9
RoI Transformer FPN [17]	69.56	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67
ICN [16]	68.16	81.36	74.30	47.70	70.32	64.89	67.82	69.98	**90.76**	79.06	78.20	53.64	62.90	67.02	64.17	50.23
RoI Transformer [17]	67.74	88.53	77.91	37.63	74.08	66.53	62.97	66.57	90.5	79.46	76.75	59.04	56.73	62.54	61.29	55.56
LH-RCNN+FPN [17], [62]	66.95	88.02	76.99	36.7	72.54	70.15	61.79	75.77	90.14	73.81	85.04	56.57	62.63	53.3	59.54	41.91
DPSRP [17]	63.89	81.18	77.42	35.48	70.41	56.74	50.42	53.56	89.97	79.68	76.48	61.99	59.94	53.34	64.04	47.76

* DOTA includes 15 categories, *i.e.,* plane (PL), baseball diamond (BD), bridge, ground track field (GTF), small vehicle (SV), large vehicle (LV), ship, tennis court (TC), basketball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (RA), harbor, swimming pool (SP) and helicopter (HC).

The FR-Est shows performance of 72.71% $mAP_{0.5}$ and outperforms the other methods with significant improvement. Noting that the methods (e.g., CAD-Net, rotated cascade and regional (RCR)-CNN, and feature fusion architecture (FFA) [6], [27], [28]) utilize FPN to improve the object detector. Here, the FR-Est without FPN achieves improvement of 2.8/3.3% $mAP_{0.5}$ over the CAD-Net/RCR-CNN with FPN. Using ResNet-101 and FPN, the FR-Est (FPN) achieves significant boost of 4.3/4.7% $mAP_{0.5}$ over the CAD-Net/RCR-CNN. Moreover, the FR-Est with ResNet-50 (Table VI) also outperforms the CAD-Net with ResNet-101-FPN (70.07 mAP vs. 69.9 mAP). It shows that the point-based estimator behaves superior performance than the regression-based method even if the latter one uses a deeper backbone.

The methods (e.g., FFA, RoI transformer, and image cascade network (ICN) [6], [16], [17]) utilize several tricks (data augmentation, multiscale testing and multiscale training) to promote the object detector. We also conduct the multiscale training and testing on the point-based estimator[8]. The FR-Est-FPN* shows superior performance of 77.62% $mAP_{0.5}$ than most of the other methods [2], [6], [17]. Moreover, the FR-Est without multiscale strategies also outperforms the RoI Transformer with tricks (72.71 mAP vs. 69.56 mAP). Meanwhile, the experimental result on HRSC2016 testing set is shown in Table IX. Our method outperforms the previous methods [14], [17], [48], [51] and achieves state-of-the-art performance. The comparison result verifies the effectiveness of our method on arbitrary-oriented object detection in aerial images.

G. Visualization of the Proposed Method

Point-based estimator shows promising performance on object detection in aerial images, as shown in Fig. 13.

[8]The implementation detail is reported in Appendix V.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:47:08 UTC from IEEE Xplore. Restrictions apply.

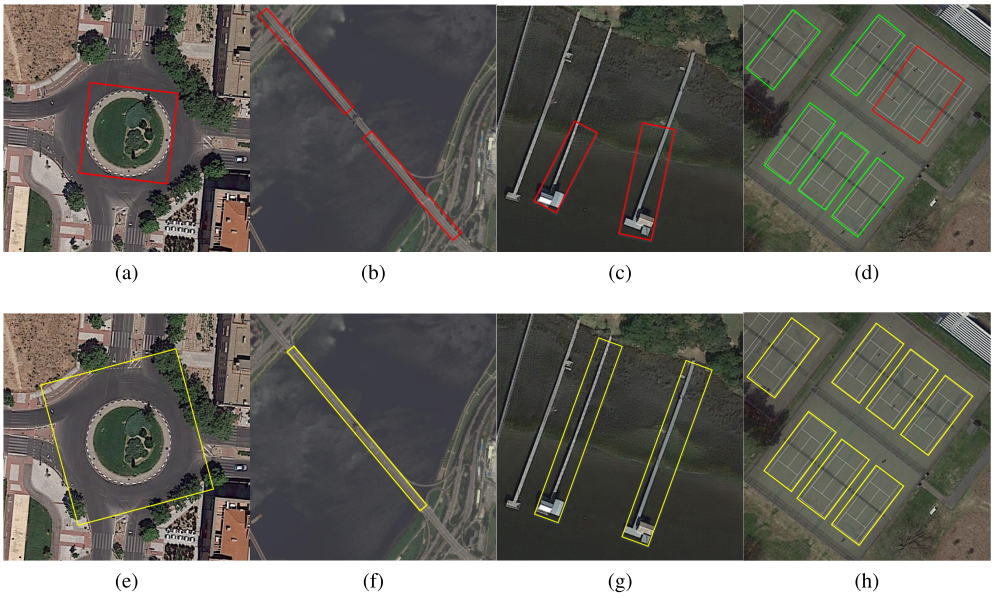


Fig. 12. Failure cases for specific categories of DOTA data set. The first row labels failure detections with red rectangles. The second row labels ground-truth objects with yellow boxes.

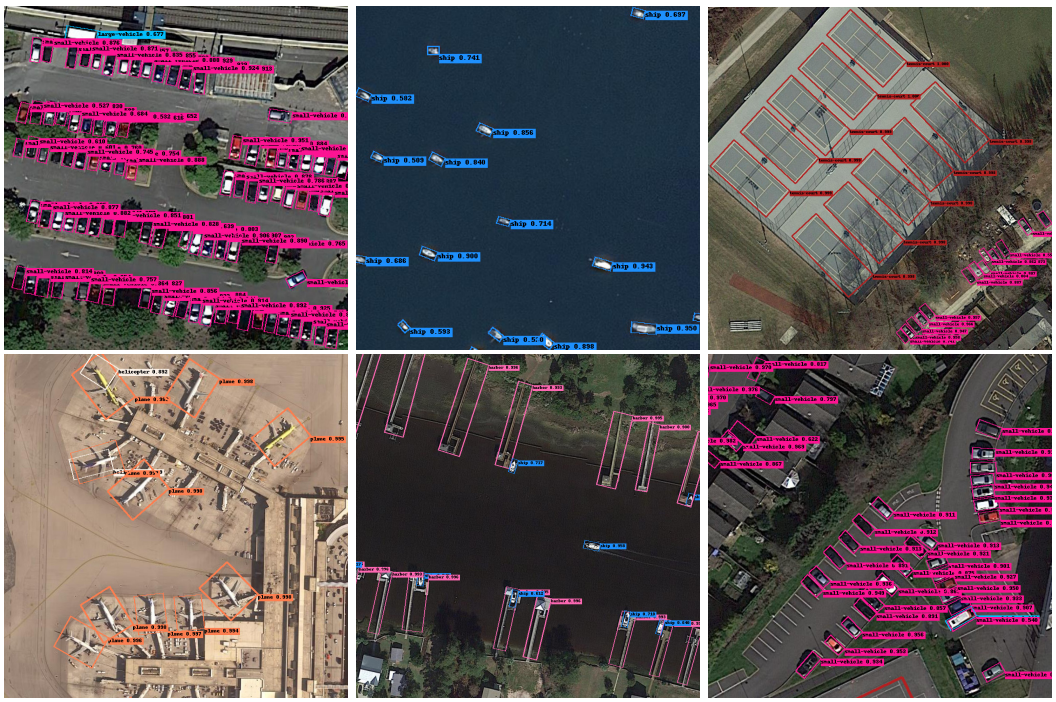


Fig. 13. Detection results of our networks on the DOTA testing set. The results show that our network has excellent capability for detecting objects with arbitrary-orientation. (The picture is best viewed when zoomed-in view).

However, there still remains a few difficult cases for our approach. Fig. 12 illustrates several typical failure cases. As shown in Fig. 12(a)/12(e), the roundabout instance does not have a clear orientation and the boundary of object has an ambiguous appearance. The object detector only captures the main body of the object and fails to localize the boundary of object. Meanwhile, the detection box has a misaligned angle with respect to the annotation. As shown in Fig. 12(b) and (f), the bridge has a long size and shares similar texture along the body. The object detector treats a part of the bridge as the entirety of instance. The model has limited receptive filed to capture the whole instance and captures insufficient clues to localize the endpoints of the bridge. Fig. 12(c) and (g), the harbors have a long and narrow corridor. The model detects the salient part of harbors and fails to localize the whole corridor in the difficult cases. In the failure cases of Fig. 12(d) and (h), the objects have similar color with the BG and the texture of instance is blurred. The object treats two adjacent objects as one instance. Overall, the object detector fails to handle few difficult cases including discriminative

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:47:08 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
PERFORMANCE OF RELEASED METHODS ON HRSC2016 TESTING SET

Method	mAP%
CP [14]	55.7
BL2 [14]	69.6
RC1 [14]	75.7
RC2 [14]	75.7
R^2PN [47]	79.6
RRD [63]	84.3
SARD [51]	85.4
RoI Transformer [17]	86.2
Ours	89.7

texture, narrow-long-shape, and ambiguous boundary. In the aerial images, the objects may have insufficient visual clues and suffer from the noises of the sensors. The model fails to construct discriminative representations for hard cases. Aggregating contextual information to the object detector can release the issue of ambiguous appearance [28], [65]–[67]. We are looking forward to combining point-based estimator with context methods.

V. CONCLUSION

In this article, we propose a novel method to handle the arbitrary-oriented object detection in aerial images. The detector exploits the point-based representation and FCN to utilize the spatial information explicitly. The point-based estimator decouples object detection task and develops specific module for each subtask. We conduct ablation experiments on the DOTA data sets and demonstrate the effectiveness of our works. Experiments showed that: 1) Decoupling the object detector into task-specific pathways removes the dilemma of feature construction. 2) On the instance localization, the point-based method and the FCN architecture can make full use of spatial information and improve the accuracy of the detection box. 3) The coarse-to-fine manner improves the precision of key-point-localization. The accurate position of points is benefit for the estimation of OBB consequently. Extensive experiments demonstrate that the point-based estimator brings consistent improvement and achieves state-of-the-art performance.

Discussion Section IV-D5 shows that the point-based estimator is complementary to the developments on the backbone and multiscale architecture. Recent studies [24], [68], [69] bring new inceptions on the design of multiscale feature. In the future research, we are looking forward to combining the method with the multiscale architecture. The pathway of object recognition only captures the feature inside the OBB, and less contextual information could be used. The works [13], [28] point out that leveraging context information can enhance the regional feature and release ambiguity of object appearance. Interpolating context into the information flow is also a meaningful research topic in our future work.

APPENDIX A
EXPLORE THE UPPER-BOUND OF POINT-BASED-ESTIMATOR

To explore the upper-bound of the point-based-estimator, we conduct experiments on the DOTA data set [2] in this section. We still concentrate on the OBB detection task in DOTA [2].

A. Methods for Improvement

Additional methods are involved to enhance the object detector, and the details are listed as follows:

1) *Multiscale Training:* As mentioned in Section IV-B, the aerial images in DOTA have large sizes (800×800 pixels to 4000×4000 pixels). During training, a window of 1000×1000 operates on the raw images to crop patches and resize the patches with aspect ratios of (0.5, 1.0). The detector is trained on the patches instead of raw images. Multiscale training can promote the capability of a model on objects of diverse scales [20], [70].
2) *Multiscale Testing:* As mentioned in Section IV-B, a window of 1000×1000 slides on the input image to crop patches during inference. Then, an image pyramid is built on the patch with ratios of (0.5, 1.0), and the object detector processes each level of pyramid. The outputs of patches are merged to make up the detection results of whole images. NMS with IoU threshold of 0.3 is utilized to filter overlapping boxes. Multiscale testing can promote the capability of model on object of diverse scales [20], [70].
3) *Enhanced Backbone:* Light-head (LH) R-CNN proposed utilizes the large-separate-convolution module to provide larger receptive field for the network. This design can enhance the feature representation of the backbone. In our method, the large-separate-convolution module is inserted into the end of ResNet backbone and feeds feature maps to the following modules. Deformable convolution is proposed to enhance the transformation modeling capability of convolutional networks [56]. In our method, the conv5-*x* stage of ResNet is equipped with deformable convolutions following previous works [46], [56].

B. Implementation Details of Models

In addition to the FR-Est and FR-Est-FPN (Section IV-F and Table VIII), several variants of the point-based estimator are equipped with advanced modules. ResNet-101 [46] serves as the backbone, and the details of detectors are listed as follows:

1) *FR-Est-LH:* this model develops the FR-Est by adding the large-separate-convolution to the ResNet backbone.
2) *FR-Est-dc:* it promotes the FR-Est by adding the deformable convolutions to the conv5-*x* stage of ResNet.
3) *FR-Est-multiple scale training (MT):* it inherits the network architecture from the FR-Est-FPN. The backbone stacks ResNet and FPN. In addition, multiscale training is involved in the training phase.
4) *FR-Est-multiple scale training and multiple scale testing (MST):* it shares the same design with the FR-Est-FPN. Multiscale training and multiscale testing strategy are involved.
5) *FR-Est-MST-deformable convolutional network (DCN):* it develops the FR-Est-FPN by equipping the conv5-*x* stage of ResNet with deformable convolutions. Multiscale training & testing is involved in the training and inference.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:47:08 UTC from IEEE Xplore. Restrictions apply.

TABLE X
PERFORMANCE OF ENHANCED POINT-BASED ESTIMATOR ON THE DOTA TESTING

Method	mAP%	Plane	BD	Bridge	GTF	SV	LV	Ship	TC	BC	ST	SBF	RA	Harbor	SP	HC
FR-Est-MST-DCN	**78.49**	89.78	**85.21**	**55.4**	**77.7**	**80.26**	**83.78**	**87.59**	90.81	**87.66**	**86.93**	**65.6**	**68.74**	**71.64**	**79.99**	**66.2**
FR-Est-MST	77.62	89.77	85.16	55.32	76.47	80.08	83.26	87.53	90.61	87.36	86.89	65.09	68.48	71.63	79.6	66.1
FR-Est-MT	76.02	89.71	83.3	52.96	73.79	77.02	80.67	87.07	89.97	86.06	85.73	63.0	67.6	71.18	74.57	62.67
FFA++ [6]	75.7	**90.1**	82.7	54.2	75.2	71.0	79.9	83.5	90.7	83.9	84.6	61.2	68.0	70.7	76.0	63.7
FR-Est-DC	74.32	89.63	81.39	50.6	70.46	73.61	78.33	86.47	90.33	84.2	83.63	60.74	66.69	70.65	67.46	60.64
FR-Est-FPN	74.20	89.63	81.17	50.44	70.19	73.52	77.98	86.44	90.82	84.13	83.56	60.64	66.59	70.59	66.72	60.55
FR-Est-LH	73.95	89.58	81.8	49.76	72.67	71.15	75.03	86.38	88.85	84.18	83.86	61.78	67.09	68.96	68.71	59.42
FR-Est	72.71	89.49	80.32	48.45	69.70	68.87	72.57	85.97	90.81	83.24	83.13	60.63	66.43	68.16	65.05	57.77
FFA-3+ [6]	71.8	89.6	76.4	48.2	58.9	67.2	76.5	81.4	90.1	83.3	83.4	55.7	60.2	73.2	66.7	64.9
CAD-Net [28]	69.9	87.8	82.4	49.4	73.5	71.1	63.5	76.7	90.9	79.2	73.3	48.4	60.9	62.0	67.0	62.2
RoI Transformer^{FPN} [17]	69.56	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67
RCR-CNN [27]	69.47	89.93	77.66	44.01	62.77	67.00	69.80	79.57	90.85	80.19	78.64	58.82	56.50	68.57	63.64	53.03
ICN [16]	68.16	81.36	74.30	47.70	70.32	64.89	67.82	69.98	90.76	79.06	78.20	53.64	62.90	67.02	64.17	50.23
RoI Transformer [17]	67.74	88.53	77.91	37.63	74.08	66.53	62.97	66.57	90.5	79.46	76.75	59.04	56.73	62.54	61.29	55.56
CAD-Net [28]	67.4	88.3	71.7	51.4	66.5	72.4	64.5	76.7	90.8	77.3	74.2	45.9	60.2	65.7	56.7	48.3
LosSLPR [27]	66.95	88.92	76.88	42.13	57.18	67.21	64.98	78.36	**90.86**	79.22	78.45	51.94	56.25	57.79	61.87	52.08

TABLE XI
PERFORMANCE OF DETECTOR UNDER DIFFERENT HYPERPARAMETER

λ	0.01	0.1	1.0	10.0	100.0
mAP%	67.23	68.09	68.37	68.41	67.52

C. Experimental Result and Analysis

The experimental result is shown in Table X. The vanilla FR-Est shows the performance of 72.71% $\mathrm{mAP}_{0.5}$. After adding the large-convolution module, the FR-Est-LH achieves the improvement of 1.2% $\mathrm{mAP}_{0.5}$ over the vanilla FR-Est. The FR-Est-dc achieves the improvement of 1.5% $\mathrm{mAP}_{0.5}$ over the vanilla FR-Est via the deformable convolution. The point-based-estimator shows consistent improvement via enhancing the capability of the backbone. The result shows that our method is complementary to the developments on backbone design, which is consistent with the experimental result in Section IV-D5. Multiscale training improves the FR-Est-MT with 1.8% $\mathrm{mAP}_{0.5}$ over the FR-Est-FPN. The FR-Est-MST shows consistent improvement of 1.6% $\mathrm{mAP}_{0.5}$ over the FR-Est-MT. With all the advanced methods, the FR-Est-MST-DCN shows the best performance of 78.49% $\mathrm{mAP}_{0.5}$. The experimental result shows that the point-based-estimator can achieve higher detection accuracy via the developments on training strategy, inference pipeline, and the network architecture design. The point-based-estimator was used as the framework to climb the top-5-rank of DOTA leader-board[9].

APPENDIX B
HYPERPARAMETER IN THE LOSS FUNCTION

The loss function in 20 sums the classification loss and localization loss. This function can be instantiated with weighted-sum operation

$$L_{\mathrm{kp}} = L_{\mathrm{cls}} + \lambda L_{\mathrm{pt}}. \tag{23}$$

Here, the hyperparameter λ controls the balance between the classification loss L_{cls} and point localization loss L_{pt}. With $\lambda = 1$, 23 is identical to 20. To investigate the training loss, we conduct experiments on FR-estimator (EST) by varying the hyperparameter λ. Table XI reports the evaluation result on the DOTA validation set. The model is insensitive to the values of λ in a wide range.

[9]The DOTA leader-board is presented at https://captain-whu.github.io/DOTA/index.html. Our team is named *changzhonghan* in DOTA task1 leader-board.

REFERENCES

[1] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.

[2] G.-S. Xia *et al.*, "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[3] X. X. Zhu *et al.*, "Deep learning in remote sensing: A comprehensive review and list of resources," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 8–36, Dec. 2017.

[4] L. Zhang, L. Zhang, and B. Du, "Deep learning for remote sensing data: A technical tutorial on the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 22–40, Jun. 2016.

[5] G. Cheng, J. Han, P. Zhou, and L. Guo, "Multi-class geospatial object detection and geographic image classification based on collection of part detectors," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 119–132, Dec. 2014. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S0924271614002524

[6] K. Fu, Z. Chang, Y. Zhang, G. Xu, K. Zhang, and X. Sun, "Rotation-aware and multi-scale convolutional neural network for object detection in remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 161, pp. 294–308, Mar. 2020. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S0924271620300319

[7] X. Zhang *et al.*, "Geospatial object detection on high resolution remote sensing imagery based on double multi-scale feature pyramid network," *Remote Sens.*, vol. 11, no. 7, p. 755, Mar. 2019.

[8] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[9] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, F. Pereira, C. J. C. Burges, L. Bottou, and K. Q. Weinberger, Eds. Red Hook, NY, USA: Curran Associates, 2012, pp. 1097–1105. [Online]. Available: http://papers.nips.cc/paper/4824-imagenet-classification-with-deep-convolutional-neural-networks.pdf

[10] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Represent.*, 2015, pp. 1–14.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:47:08 UTC from IEEE Xplore. Restrictions apply.

[11] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[12] Z. Deng, H. Sun, S. Zhou, J. Zhao, and H. Zou, "Toward fast and accurate vehicle detection in aerial images using coupled region-based convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 8, pp. 3652–3664, Aug. 2017.

[13] K. Li, G. Cheng, S. Bu, and X. You, "Rotation-insensitive and context-augmented object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2337–2348, Apr. 2018.

[14] Z. Liu, J. Hu, L. Weng, and Y. Yang, "Rotated region based CNN for ship detection," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 900–904.

[15] J. Ding *et al.*, "ICPR2018 contest on object detection in aerial images (ODAI-18)," in *Proc. 24th Int. Conf. Pattern Recognit. (ICPR)*, Aug. 2018, pp. 1–6.

[16] S. M. Azimi, E. Vig, R. Bahmanyar, M. Körner, and P. Reinartz, "Towards multi-class object detection in unconstrained remote sensing imagery," in *Proc. 14th Asian Conf. Comput. Vis., (ACCV)*, Perth, Australia, Dec. 2018, pp. 150–165, doi: 10.1007/978-3-030-20893-6_10

[17] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for detecting oriented objects in aerial images," *CoRR*, vol. abs/1812.00155, pp. 1–19, Dec. 2018.

[18] J. Koo, J. Seo, S. Jeon, J. Choe, and T. Jeon, "RBox-CNN: Rotated bounding box based CNN for ship detection in remote sensing image," in *Proc. 26th ACM SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst.*, Nov. 2018, pp. 420–423.

[19] L. Liu, Z. Pan, and B. Lei, "Learning a rotation invariant detector with rotatable bounding box," *CoRR*, vol. abs/1711.09405, pp. 1–9, Nov. 2017.

[20] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 2, pp. 386–397, Feb. 2020.

[21] C. Szegedy *et al.*, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.

[22] P. O. Pinheiro, T. Lin, R. Collobert, and P. Dollár, "Learning to refine object segments," in *Proc. 14th Eur. Conf. Comput. Vis. (ECCV)*, Amsterdam, The Netherlands, Oct. 2016, pp. 75–91, doi: 10.1007/978-3-319-46448-0_5.

[23] G. Lin, F. Liu, A. Milan, C. Shen, and I. Reid, "RefineNet: Multi-path refinement networks for dense prediction," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 5, pp. 1228–1242, May 2020.

[24] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.

[25] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2921–2929.

[26] X. Lu, B. Li, Y. Yue, Q. Li, and J. Yan, "Grid R-CNN," 2018, *arXiv:1811.12030*. [Online]. Available: https://arxiv.org/abs/1811.12030

[27] Y. Zhu, C. Ma, and J. Du, "Rotated cascade R-CNN: A shape robust detector with coordinate regression," *Pattern Recognit.*, vol. 96, Dec. 2019, Art. no. 106964. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S0031320319302614

[28] G. Zhang, S. Lu, and W. Zhang, "CAD-net: A context-aware detection network for objects in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10015–10024, Dec. 2019.

[29] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[30] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[31] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 580–587.

[32] S. Ren, K. He, R. Girshick, X. Zhang, and J. Sun, "Object detection networks on convolutional feature maps," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 7, pp. 1476–1481, Jul. 2017.

[33] Y. Wu *et al.*, "Rethinking classification and localization for object detection," 2019, *arXiv:1904.06493*. [Online]. Available: http://arxiv.org/abs/1904.06493

[34] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[35] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[36] H. Law and J. Deng, "Cornernet: Detecting objects as paired keypoints," in *Computer Vision—ECCV*, V. Ferrari, M. Hebert, C. Sminchisescu, and Y. Weiss, Eds. Cham, Switzerland: Springer, 2018, pp. 765–781.

[37] X. Zhou, J. Zhuo, and P. Krähenbühl, "Bottom-up object detection by grouping extreme and center points," 2019, *arXiv:1901.08043*. [Online]. Available: http://arxiv.org/abs/1901.08043

[38] A. Newell, K. Yang, and J. Deng, "Stacked hourglass networks for human pose estimation," in *Computer Vision—ECCV*, B. Leibe, J. Matas, N. Sebe, and M. Welling, Eds. Cham, Switzerland: Springer, 2016, pp. 483–499.

[39] M. Andriluka, L. Pishchulin, P. Gehler, and B. Schiele, "2D human pose estimation: New benchmark and state of the art analysis," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 3686–3693.

[40] T.-Y. Lin *et al.*, "Microsoft COCO: Common objects in context," in *Computer Vision—ECCV*, D. Fleet, T. Pajdla, B. Schiele, and T. Tuytelaars, Eds. Cham, Switzerland: Springer, 2014, pp. 740–755.

[41] Z. Cao, T. Simon, S.-E. Wei, and Y. Sheikh, "Realtime multi-person 2D pose estimation using part affinity fields," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1302–1310.

[42] H.-S. Fang, S. Xie, Y.-W. Tai, and C. Lu, "RMPE: Regional multi-person pose estimation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2353–2362.

[43] G. Papandreou *et al.*, "Towards accurate multi-person pose estimation in the wild," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3711–3719.

[44] Y. Zhang, Y. Yuan, Y. Feng, and X. Lu, "Hierarchical and robust convolutional neural network for very high-resolution remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5535–5548, Aug. 2019.

[45] J. Jiao *et al.*, "A densely connected end-to-end neural network for multiscale and multiscene SAR ship detection," *IEEE Access*, vol. 6, pp. 20881–20892, 2018.

[46] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[47] Z. Zhang, W. Guo, S. Zhu, and W. Yu, "Toward arbitrary-oriented ship detection with rotated region proposal and discrimination networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 11, pp. 1745–1749, Nov. 2018.

[48] Z. Liu, H. Wang, L. Weng, and Y. Yang, "Ship rotated bounding box space for ship extraction from high-resolution optical satellite images with complex backgrounds," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 8, pp. 1074–1078, Aug. 2016.

[49] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9627–9636.

[50] J. Ma *et al.*, "Arbitrary-oriented scene text detection via rotation proposals," *IEEE Trans. Multimedia*, vol. 20, no. 11, pp. 3111–3122, Nov. 2018.

[51] Y. Wang, Y. Zhang, Y. Zhang, L. Zhao, X. Sun, and Z. Guo, "SARD: Towards scale-aware rotated object detection in aerial imagery," *IEEE Access*, vol. 7, pp. 173855–173865, 2019.

[52] Y. Jiang *et al.*, "R2CNN: Rotational region CNN for orientation robust scene text detection," 2017, *arXiv:1706.09579*. [Online]. Available: https://arxiv.org/abs/1706.09579

[53] Y. Wu, A. Kirillov, F. Massa, W.-Y. Lo, and R. Girshick. (2019). *Detectron2*. [Online]. Available: https://github.com/facebookresearch/detectron2

[54] F. Massa and R. Girshick. (2018). *Maskrnn-Benchmark: Fast, Modular Reference Implementation of Instance Segmentation and Object Detection Algorithms in PyTorch*. [Online]. Available: https://github.com/facebookresearch/maskrcnn-benchmark

[55] T.-Y. Lin, P. Dollar, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

[56] J. Dai *et al.*, "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 764–773.

[57] Y. LeCun *et al.*, "Backpropagation applied to handwritten zip code recognition," *Neural Comput.*, vol. 1, no. 4, pp. 541–551, Dec. 1989.

[58] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, Jun. 2010, doi: 10.1007/s11263-009-0275-4.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:47:08 UTC from IEEE Xplore. Restrictions apply.

[59] M. Everingham, S. M. A. Eslami, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes challenge: A retrospective," *Int. J. Comput. Vis.*, vol. 111, no. 1, pp. 98–136, Jan. 2015, doi: 10.1007/s11263-014-0733-5.
[60] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.
[61] X. Yang, H. Sun, X. Sun, M. Yan, Z. Guo, and K. Fu, "Position detection and direction prediction for arbitrary-oriented ships via multitask rotation region convolutional neural network," 2018, *arXiv:1806.04828*. [Online]. Available: http://arxiv.org/abs/1806.04828
[62] Z. Li, C. Peng, G. Yu, X. Zhang, Y. Deng, and J. Sun, "Light-head R-CNN: In defense of two-stage object detector," 2017, *arXiv:1711.07264*. [Online]. Available: http://arxiv.org/abs/1711.07264
[63] M. Liao, Z. Zhu, B. Shi, G.-S. Xia, and X. Bai, "Rotation-sensitive regression for oriented scene text detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5909–5918.
[64] D. Hoiem, Y. Chodpathumwan, and Q. Dai, "Diagnosing error in object detectors," in *Proc. 12th Eur. Conf. Comput. Vis.*, Florence, Italy, Oct. 2012, pp. 340–353, doi: 10.1007/978-3-642-33712-3_25.
[65] J. He, Z. Deng, L. Zhou, Y. Wang, and Y. Qiao, "Adaptive pyramid context network for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7519–7528.
[66] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239.
[67] Y. Feng, W. Diao, X. Sun, M. Yan, and X. Gao, "Towards automated ship detection and category recognition from high-resolution aerial images," *Remote Sens.*, vol. 11, no. 16, p. 1901, Aug. 2019. [Online]. Available: https://www.mdpi.com/2072-4292/11/16/1901
[68] T. Kong, F. Sun, W. Huang, and H. Liu, "Deep feature pyramid reconfiguration for object detection," in *Proc. 15th Eur. Conf.*, Munich, Germany, Sep. 2018, pp. 172–188, doi: 10.1007/978-3-030-01228-1_11.
[69] G. Ghiasi, T.-Y. Lin, R. Pang, and Q. V. Le, "NAS-FPN: Learning scalable feature pyramid architecture for object detection," 2019, *arXiv:1904.07392*. [Online]. Available: http://arxiv.org/abs/1904.07392
[70] W. Liu *et al.*, "SSD: Single shot multibox detector," in *Computer Vision—ECCV*, B. Leibe, J. Matas, N. Sebe, and M. Welling, Eds. Cham, Switzerland: Springer, 2016, pp. 21–37.



Kun Fu received the B.Sc., M.Sc., and Ph.D. degrees from the National University of Defense Technology, Changsha, China, in 1995, 1999, and 2002, respectively.

He is a Professor with Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include computer vision, remote sensing image understanding, geospatial data mining, and visualization.



Zhonghan Chang received the B.Sc. degree from Tianjin University, Tianjin, China, in 2016. He is pursuing the Ph.D. degree with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

His research interests include computer vision, pattern recognition, and remote sensing image processing, especially on object detection.



Yue Zhang (Member, IEEE) received the B.E. degree in electronic engineering from Northwestern Polytechnical University, Xi'an, China, in 2012, and the Ph.D. degree from the University of Chinese Academy of Sciences, Beijing, China, in 2017.

He is an Assistant Professor with Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His research interests include the analysis of optical and synthetic aperture radar remote sensing images.



Xian Sun (Senior Member, IEEE) received the B.Sc. degree from Beihang University, Beijing, China, in 2004, and the M.Sc. and Ph.D. degrees from Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, in 2006 and 2009, respectively.

He is a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision and remote-sensing image understanding.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:47:08 UTC from IEEE Xplore. Restrictions apply.